

INTERNATIONAL STANDARD

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**Programmable controllers –
Part 3: Programming languages**

**Automates programmables –
Partie 3: Langages de programmation**



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CONTENTS

FOREWORD.....	7
1 Scope.....	9
2 Normative references.....	9
3 Terms and definitions.....	9
4 Architectural models.....	17
4.1 Software model.....	17
4.2 Communication model.....	19
4.3 Programming model.....	20
5 Compliance.....	22
5.1 General.....	22
5.2 Feature tables.....	22
5.3 Implementer's compliance statement.....	22
6 Common elements.....	24
6.1 Use of printed characters.....	24
6.1.1 Character set.....	24
6.1.2 Identifiers.....	24
6.1.3 Keywords.....	25
6.1.4 Use of white space.....	25
6.1.5 Comments.....	25
6.2 Pragma.....	26
6.3 Literals – External representation of data.....	26
6.3.1 General.....	26
6.3.2 Numeric literals and string literals.....	26
6.3.3 Character string literals.....	28
6.3.4 Duration literal.....	31
6.3.5 Date and time of day literal.....	31
6.4 Data types.....	32
6.4.1 General.....	32
6.4.2 Elementary data types (BOOL, INT, REAL, STRING, etc.).....	32
6.4.3 Generic data types.....	36
6.4.4 User-defined data types.....	37
6.5 Variables.....	51
6.5.1 Declaration and initialization of variables.....	51
6.5.2 Variable sections.....	54
6.5.3 Variable length <code>ARRAY</code> variables.....	57
6.5.4 Constant variables.....	59
6.5.5 Directly represented variables (%).....	59
6.5.6 Retentive variables (RETAIN, NON_RETAIN).....	62
6.6 Program organization units (POUs).....	63
6.6.1 Common features for POU's.....	63
6.6.2 Functions.....	77
6.6.3 Function blocks.....	113
6.6.4 Programs.....	132
6.6.5 Classes.....	134
6.6.6 Interface.....	155
6.6.7 Object-oriented features for function block types.....	165

6.6.8	Polymorphism.....	171
6.7	Sequential function chart (SFC) elements.....	174
6.7.1	General.....	174
6.7.2	Steps.....	175
6.7.3	Transitions	176
6.7.4	Actions	179
6.7.5	Rules of evolution	187
6.8	Configuration elements	195
6.8.1	General.....	195
6.8.2	Tasks	200
6.9	Synchronization of concurrent execution	205
6.9.1	General.....	205
6.9.2	Mutex	206
6.9.3	Object-oriented mutex	208
6.9.4	Semaphore.....	209
6.9.5	Object-oriented semaphore	211
6.10	Namespaces	211
6.10.1	General.....	211
6.10.2	Declaration.....	212
6.10.3	Usage.....	217
6.10.4	Namespace directive USING	218
7	Textual languages	221
7.1	Common elements.....	221
7.2	Structured text (ST)	221
7.2.1	General.....	221
7.2.2	Expressions	221
7.2.3	Statements.....	223
8	Graphical languages	229
8.1	Common elements.....	229
8.1.1	General.....	229
8.1.2	Representation of variables and instances.....	229
8.1.3	Representation of lines and blocks	231
8.1.4	Direction of flow in networks	232
8.1.5	Evaluation of networks	233
8.1.6	Execution control elements	234
8.2	Ladder diagram (LD).....	235
8.2.1	General.....	235
8.2.2	Power rails	236
8.2.3	Link elements and states	236
8.2.4	Contacts	236
8.2.5	Coils.....	238
8.2.6	Functions and function blocks	239
8.2.7	Order of network evaluation	239
8.3	Function block diagram (FBD).....	240
8.3.1	General.....	240
8.3.2	Combination of elements	240
8.3.3	Order of network evaluation	240
Annex A (normative)	Formal specification of the language elements	241

Annex B (informative) List of major changes and extensions of Edition 4 of IEC 61131-3	249
B.1 General	249
B.2 New features	249
B.3 Significant changes	249
B.4 Deletions	250
B.5 Deprecations	250
Annex C (informative) Relating strings, characters, and their literals to ISO/IEC 10646	251
Bibliography	253
Figure 1 – Software model	18
Figure 2 – Communication model	20
Figure 3 – Combination of programmable controller language elements	21
Figure 4 – Implementer's compliance statement (Example)	23
Figure 5 – Hierarchy of the generic data types	37
Figure 6 – Initialization by literals and constant expressions (Rules)	38
Figure 7 – Variable declaration keywords (Summary)	54
Figure 8 – Usage of <code>VAR_GLOBAL</code> , <code>VAR_EXTERNAL</code> and <code>CONSTANT</code> (Rules)	55
Figure 9 – Conditions for the initial value of a variable (Rules)	62
Figure 10 – Formal and non-formal representation of call (Examples)	68
Figure 11 – Data type conversion rules – implicit and explicit (Summary)	72
Figure 12 – Implicit type conversions overview	73
Figure 13 – Data type conversion from string types to array of bytes	96
Figure 14 – Data type conversion from array of bytes to string types	97
Figure 15 – Usage of function block input and output parameters (Rules)	123
Figure 16 – Usage of function block input and output parameters (Illustration of rules)	124
Figure 17 – Standard timer function blocks – timing diagrams (Rules)	130
Figure 18 – Recommended implementation in pseudo code for timer	132
Figure 19 – Overview of inheritance and interface implementation	135
Figure 20 – Inheritance of classes (Illustration)	144
Figure 21 – Interface with derived classes (Illustration)	157
Figure 22 – Inheritance of interface and class (Illustration)	162
Figure 23 – Function block types with optional body and methods (Illustration)	168
Figure 24 – Inheritance of function block body with <code>SUPER()</code> (Example)	170
Figure 25 – <code>ACTION_CONTROL</code> function block – External interface (Summary)	184
Figure 26 – <code>ACTION_CONTROL</code> function block body (Summary)	185
Figure 27 – Action control (Example)	186
Figure 28 – SFC evolution (Rules)	193
Figure 29 – SFC errors (Example)	194
Figure 30 – Configuration (Example)	196
Figure 31 – <code>CONFIGURATION</code> and <code>RESOURCE</code> declaration (Example)	199
Figure 32 – The reference implementation of mutex functions	208
Figure 33 – The reference implementation of object-oriented mutex	209
Figure 34 – The reference implementation of semaphore functions	210

Figure 35 – The reference implementation of object-oriented semaphore	211
Figure 36 – Accessibility using namespaces (Rules)	214
Figure 37 – Common textual elements (Summary)	221
Figure C.1 – Relation between ISO/IEC 10646 and IEC 61131-3	251
Figure C.2 – Example of encoded strings in memory	252
Table 1 – Character set.....	24
Table 2 – Identifiers.....	24
Table 3 – Comments.....	26
Table 4 – Pragma.....	26
Table 5 – Numeric literals.....	28
Table 6 – Character string literals	29
Table 7 – Two-character combinations in character strings	30
Table 8 – Duration literals	31
Table 9 – Date and time of day literals.....	32
Table 10 – Elementary data types	33
Table 11 – Declaration of user-defined data types and initialization	38
Table 12 – Reference operations	51
Table 13 – Declaration of variables	53
Table 14 – Initialization of variables	53
Table 15 – Variable-length <code>ARRAY</code> variables	57
Table 16 – Directly represented variables	60
Table 17 – Partial access of <code>ANY_BIT</code> variables (except <code>BOOL</code>)	65
Table 18 – Execution control graphically using <code>EN</code> and <code>ENO</code>	70
Table 19 – Function declaration	79
Table 20 – Function call	81
Table 21 – Typed and overloaded functions	84
Table 22 – Data type conversion function.....	86
Table 23 – Data type conversion of numeric data types	88
Table 24 – Data type conversion of bit data types	91
Table 25 – Data type conversion of bit and numeric types.....	92
Table 26 – Data type conversion of date and time types	94
Table 27 – Data type conversion of character types	95
Table 28 – Numerical and arithmetic functions.....	98
Table 29 – Arithmetic functions.....	99
Table 30 – Bit shift functions.....	100
Table 31 – Bitwise Boolean functions	100
Table 32 – Selection functions	101
Table 33 – Comparison functions	102
Table 34 – Character string functions	103
Table 35 – Numerical functions of time and duration data types	105
Table 36 – Additional functions of time data types <code>CONCAT</code> and <code>SPLIT</code>	107
Table 37 – Function for endianness conversion	111

Table 38 – Functions of enumerated and named values data types.....	111
Table 39 – Validate functions.....	112
Table 40 – Function ASSERT	112
Table 41 – Function block type declaration.....	114
Table 42 – Function block instance declaration.....	118
Table 43 – Function block call.....	120
Table 44 – Standard bistable function blocks ^a	126
Table 45 – Standard edge detection function blocks.....	127
Table 46 – Standard counter function blocks	128
Table 47 – Standard timer function blocks.....	129
Table 48 – Program declaration	133
Table 49 – Class	136
Table 50 – Class instance declaration	138
Table 51 – Textual call of methods – Formal and non-formal parameter list.....	142
Table 52 – Interface.....	156
Table 53 – Assignment attempt.....	164
Table 54 – Object-oriented function block type	166
Table 55 – SFC step.....	176
Table 56 – SFC transition and transition condition	178
Table 57 – SFC declaration of actions	180
Table 58 – Association between step and action.....	181
Table 59 – Action block.....	182
Table 60 – Action qualifiers	182
Table 61 – Action control features	187
Table 62 – Sequence evolution – graphical	188
Table 63 – Configuration and resource declaration	198
Table 64 – Task	201
Table 65 – Mutex type and operations	207
Table 66 – Semaphore type and operations	209
Table 67 – Namespace	216
Table 68 – Nested namespace declaration options.....	217
Table 69 – Namespace directive USING.....	220
Table 70 – Operators of the ST language.....	223
Table 71 – ST language statements	224
Table 72 – Graphical execution control elements	235
Table 73 – Power rails and link elements	236
Table 74 – Contacts.....	237
Table 75 – Coils	239

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IEC 61131-3 has been prepared by subcommittee 65B: Measurement and control devices, of IEC technical committee 65: Industrial-process measurement, control and automation. It is an International Standard.

This fourth edition cancels and replaces the third edition published in 2013. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) inclusion of UTF-8 strings and their associated functions;
- b) Annex B contains a comprehensive list of features that have been added, removed or deprecated in comparison to IEC 61131-3:2013.

The text of this International Standard is based on the following documents:

Draft	Report on voting
65B/1281/FDIS	65B/1291/RVD

Full information on the voting for its approval can be found in the report on voting indicated in the above table.

The language used for the development of this International Standard is English.

This document was drafted in accordance with ISO/IEC Directives, Part 2, and developed in accordance with ISO/IEC Directives, Part 1 and ISO/IEC Directives, IEC Supplement, available at www.iec.ch/members_experts/refdocs. The main document types developed by IEC are described in greater detail at www.iec.ch/publications.

A list of all the parts in the IEC 61131 series, published under the general title *Programmable controllers* can be found on the IEC website.

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PROGRAMMABLE CONTROLLERS –

Part 3: Programming languages

1 Scope

This part of IEC 61131 specifies the syntax and semantics of programming languages for programmable controllers as defined in IEC 61131-1.

This document specifies the syntax and semantics of a unified suite of programming languages for programmable controllers (PCs). This suite consists of the textual language structured text (ST), and the graphical languages, ladder diagram (LD) and function block diagram (FBD).

An additional set of graphical and equivalent textual elements named sequential function chart (SFC) is defined for structuring the internal organization of programs and function blocks. Also, configuration elements are defined which support the installation of programmable controller programs into programmable controller systems.

In addition, features are defined which facilitate communication among programmable controllers and other components of automated systems.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61131-1, *Programmable controllers – Part 1: General information*

IEC 61131-5, *Programmable controllers – Part 5: Communications*

ISO/IEC 10646, *Information technology – Universal Coded Character Set (UCS)*

ISO/IEC 60559, *Floating-point arithmetic*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 61131-1 and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>

3.1

access path

association of a symbolic name with a variable for the purpose of open communication

3.2**action**

Boolean variable or a collection of operations to be performed, together with an associated control structure

3.3**action block**

graphical language element which utilizes a Boolean input variable to determine the value of a Boolean output variable or the enabling condition for an action, according to a predetermined control structure

3.4**aggregate**

structured collection of data objects forming a data type

3.5**array**

aggregate that consists of data objects, with identical attributes, each of which can be uniquely referenced by subscripting

3.6**assignment**

mechanism to give a value to a variable or to an aggregate

3.7**base type**

data type, function block type or class from which further types are inherited or derived

3.8**based number**

number represented in a specified base other than ten

3.9**binary coded decimal**

BCD

encoding for decimal numbers in which each digit is represented by its own binary sequence

3.10**bistable function block**

function block with two stable states controlled by one or more inputs

3.11**bit string**

data element consisting of one or more bits

3.12**bit string literal**

literal that directly represents a bit string value of data type `BOOL`, `BYTE`, `WORD`, `DWORD`, or `LWORD`

3.13**body**

set of operations of a program organization unit

3.14**call**

language construct causing the execution of a function, function block, or method

3.15**character string**

aggregate that consists of an ordered sequence of characters

3.16**character string literal**

literal that directly represents a character or character string value of data type CHAR, WCHAR, UCHAR, STRING, WSTRING or USTRING

3.17**class**

program organization unit consisting of:

- the definition of a data structure,
- a set of methods and properties to be performed upon the data structure

3.18**comment**

language construct for the inclusion of text having no impact on the execution of the program

3.19**configuration**

language element corresponding to one or multiple programmable controller system

3.20**constant**

language element which declares a data element with a fixed value

3.21**counter function block**

function block which accumulates a value for the number of changes sensed at one or more specified inputs

3.22**data type**

language element which allows a set of values together with a set of permitted operations

3.23**date and time**

date within the year and the time of day represented as a single language element

3.24**declaration**

mechanism for establishing the definition of a language element

3.25**delimiter**

character or combination of characters used to separate program language elements

3.26**derived class****extended class****child class**

class that extends another class by inheritance

3.27**directly derived data type**

data type that is declared as an alias for another data type

3.28**derived function block type**

function block type created by inheritance from another function block type or class type

3.29**direct representation**

means of representing a variable in a programmable controller program from which an implementation-specified correspondence to a physical or logical location can be determined directly

3.30**double word**

data element containing 32 bits

3.31**dynamic binding**

situation in which the implementation of a method is retrieved during runtime according to the actual type of the instance

3.32**evaluation**

process of establishing a value for an expression or a function, or for the outputs of a network or function block instance, during program execution

3.33**execution control element**

language element which controls the flow of program execution

3.34**falling edge**

change from TRUE (1) to FALSE (0) of a Boolean variable

3.35**function**

program organization unit that does not retain state and which, when executed, typically yields one result and possibly additional output variables

3.36**function block instance**

instance of a function block type

3.37**function block type**

language element consisting of:

- the definition of a data structure partitioned into input, output, and internal variables; and
- a set of operations or a set of methods and properties to be performed upon the elements of the data structure when an instance of the function block type is called

3.38**function block diagram**

network in which the nodes are function block instances, graphically represented functions or method calls, variables, literals, and labels

3.39**generic data type**

data type which represents more than one type of data

3.40**global variable**

variable whose scope is global

3.41**hierarchical addressing**

direct representation of a data element as a member of a physical or logical hierarchy

EXAMPLE A point within a module which is contained in a rack, which in turn is contained in a cubicle, etc.

3.42**identifier**

combination of letters, numbers, and underscore characters which begins with a letter or underscore and which names a language element

3.43**implementation**

product version of a PLC or the programming and debugging tool provided by the implementer

3.44**implementer**

manufacturer of the PLC or the programming and debugging tool provided to the user to program a PLC application

3.45**inheritance**

creation of a new class, function block type or interface based on an existing class, function block type or interface, respectively

3.46**initial value**

value assigned to a variable at the beginning of its lifetime

3.47**in-out variable**

variable which is used to supply a value to a program organization unit and which is additionally used to return a value from the program organization unit

3.48**input variable**

variable which is used to supply a value to a program organization unit or method

3.49**instance**

individual, named copy of the data structure associated with a function block type, class, or program type

3.50**instance name**

identifier associated with a specific instance

3.51**instantiation**

creation of an instance

3.52**integer literal**

literal which directly represents an integer value

3.53

interface

language element in the context of object-oriented programming containing a set of method and property prototypes

3.54

keyword

lexical unit reserved for use in a programming language

3.55

label

identifier associated with a network in the LD or FBD language

3.56

language element

item identified by a symbol on the left-hand side of a production rule in the formal specification

3.57

literal

lexical unit that directly represents a value

3.58

logical location

location of a hierarchically addressed variable in a schema which can bear some relation to the physical structure of the programmable controller's inputs, outputs, and memory

3.59

method

language element of a function block type or class type, which is equal to a function except that it is called on an instance and its operations can access this instance

3.60

method prototype

language element consisting only of the name and the signature of a method

3.61

named element

element of a structure which is named by its associated identifier

3.62

network

arrangement of nodes and interconnecting branches

3.63

numeric literal

literal which directly represents a numeric value, i.e. an integer literal or real literal

3.64

operation

language element that represents an elementary functionality belonging to a program organization unit or method

3.65

operand

language element on which an operation is performed